cp_10d

1

M pattern with * | 40/40

Solved

Easy

Given a number `N` , write a program to print the letter **M** of `N` rows with two **Solid Right Angled Triangles** of `N` rows using stars (`*`).

```
*                   *
* *                 * *
* * *               * * *
* * * *             * * * *
* * * * *           * * * * *
* * * * * * * * * * * * * * *
```

Note

There is a space after every star (`*`).

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be `N` rows containing stars (`*`) as the letter **M** shown in the sample output.

Explanation

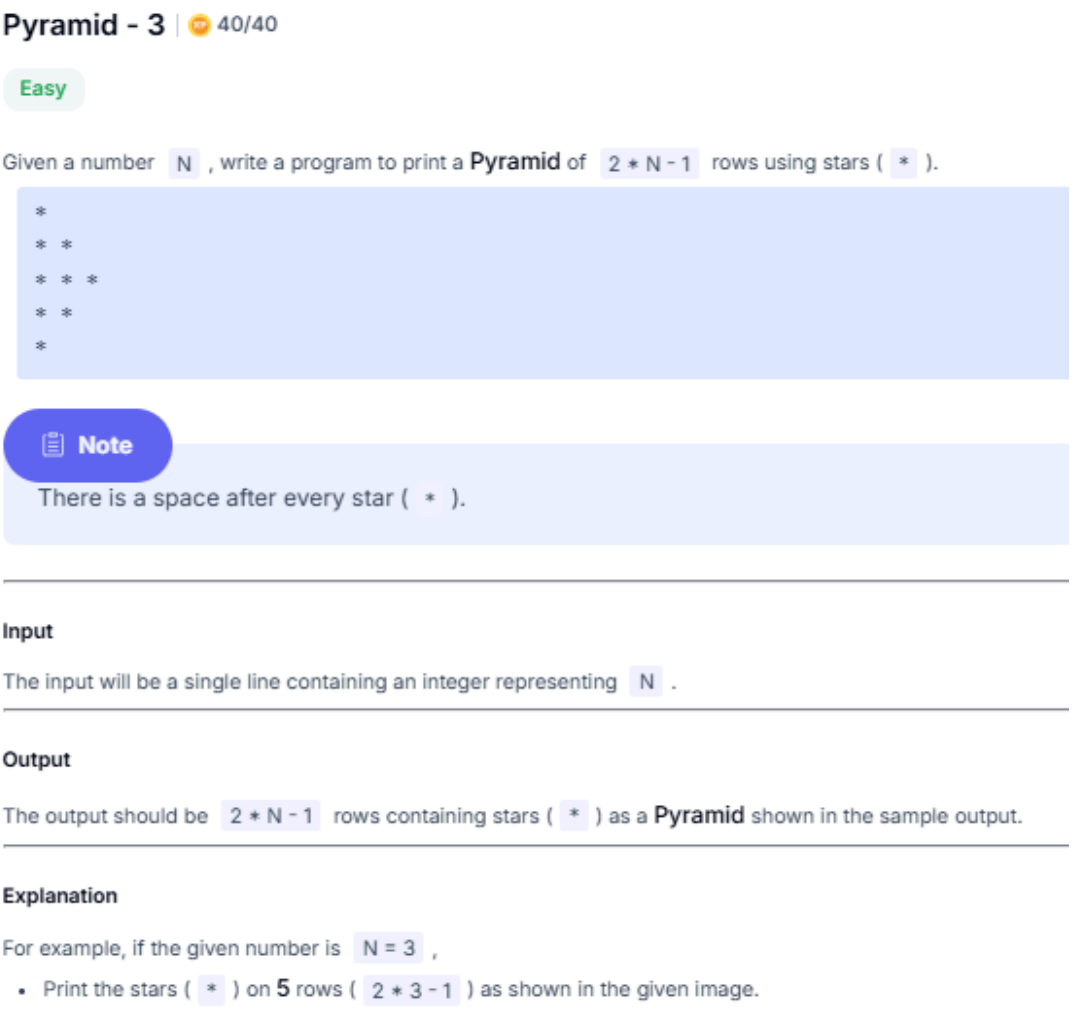
For example, if the given number is `N = 6` ,

- Print the stars (`*`) on **6** rows as shown in the given image.

code:

```
n=int(input())
for i in range(1,n+1):
    tri="* "*i
    spaces="  "*((n*2)-2)
    row=tri+spaces+tri
    n=n-1
    print(row)
```

2



Pyramid - 3 | 40/40

Easy

Given a number `N` , write a program to print a **Pyramid** of `2 * N - 1` rows using stars (`*`).

```
*
* *
* * *
* *
*
```

Note

There is a space after every star (`*`).

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be `2 * N - 1` rows containing stars (`*`) as a **Pyramid** shown in the sample output.

Explanation

For example, if the given number is `N = 3` ,

- Print the stars (`*`) on **5** rows (`2 * 3 - 1`) as shown in the given image.

code:

```
n=int(input())
for i in range(1,n*2):
    row="* "*i
    if i>n:
        row="* "*(n-1)
        n=n-1
    print(row)
```

3

Pyramid - 4 | 40/40

Easy

Given a number `N` , write a program to print a **Pyramid** of `2 * N - 1` rows using numbers.

```
1
2 2
3 3 3
4 4 4 4
5 5 5 5 5
4 4 4 4
3 3 3
2 2
1
```

Note

There is a space after every number.

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be `2 * N - 1` rows containing numbers as a **Pyramid** shown in the sample output.

Explanation

For example, if the given number is `N = 5` ,

- Print the numbers (`1`) on **9** rows (`2 * 5 - 1 = 9`) as shown in the given image.

code:

```
n=int(input())
for i in range(1,n*2):
    str_i=str(i)+" "
    row=str_i*i
    if i>n:
        n=n-1
        str_n=str(n)+" "
        row=str_n*(n)
    print(row)
```

4

Pyramid - 5 | 40/40

Solved

Easy

Given a number `N` , write a program to print a **Pyramid** of `2 * N - 1` rows using pluses (`+`) and hashes (`#`).
The last column of the pyramid contains hashes (`#`), and the remaining columns contain pluses (`+`).



Note

There is a space after every plus (`+`) and every hash (`#`).

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be `2 * N - 1` rows containing pluses (`+`) and hashes (`#`) as a **Pyramid** shown in the sample output.

Explanation

For example, if the given number is `N = 5` ,

- Print pluses (`+`) and hashes (`#`) on **9** rows (`(2 * 5) - 1`) as shown in the given image.

code:

```
n=int(input())
n1=n
for i in range(n):
    space="  "*(n-1)
    plus="+ "*i
    row=space+plus+"# "
    print(row)
    n=n-1
for i in range(n1-1):
    space="  "*(i+1)
```

```
    plus="+ "*(n1-2)
    row=space+plus+"# "
    print(row)
    n1=n1-1
```

5

Butterfly | 40/40

Easy

Given a number `N` , write a program to print a **Butterfly** of `(2 * N) - 1` rows using stars (`*`).

```
*         *
* *     * *
* * * * * *
* *     * *
*         *
```

Note

There is a space after every star (`*`).

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be `(2 * N) - 1` rows containing stars (`*`) as a **Butterfly** shown in the sample output.

Explanation

For example, if the given number is `N = 4` ,

- Print the stars (`*`) on **7** rows `((2 * 4) - 1)` as shown in the given image.

code:

```
n=int(input())
n1=n
for i in range(1,n+1):
    star="* "*i
    space="  "*(2*(n-1))
```

```
    row=star+space+star
    print(row)
    n=n-1
for i in range(1,n1):
   star="* "*(n1-1)
   space="  "*(2*(i))
   row=star+space+star
   print(row)
   n1=n1-1
```

6



Pyramid - 6 | 40/40 ● Solved

Easy

Given a number `N` , write a program to print a **Pyramid** of `N` rows using stars (`*`).

```
  *
 * *
* * *
```

Note

There is a space after every star (`*`).

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be `N` rows containing stars (`*`) as a **Pyramid** shown in the sample output.

Explanation

For example, if the given number is `N = 4` ,

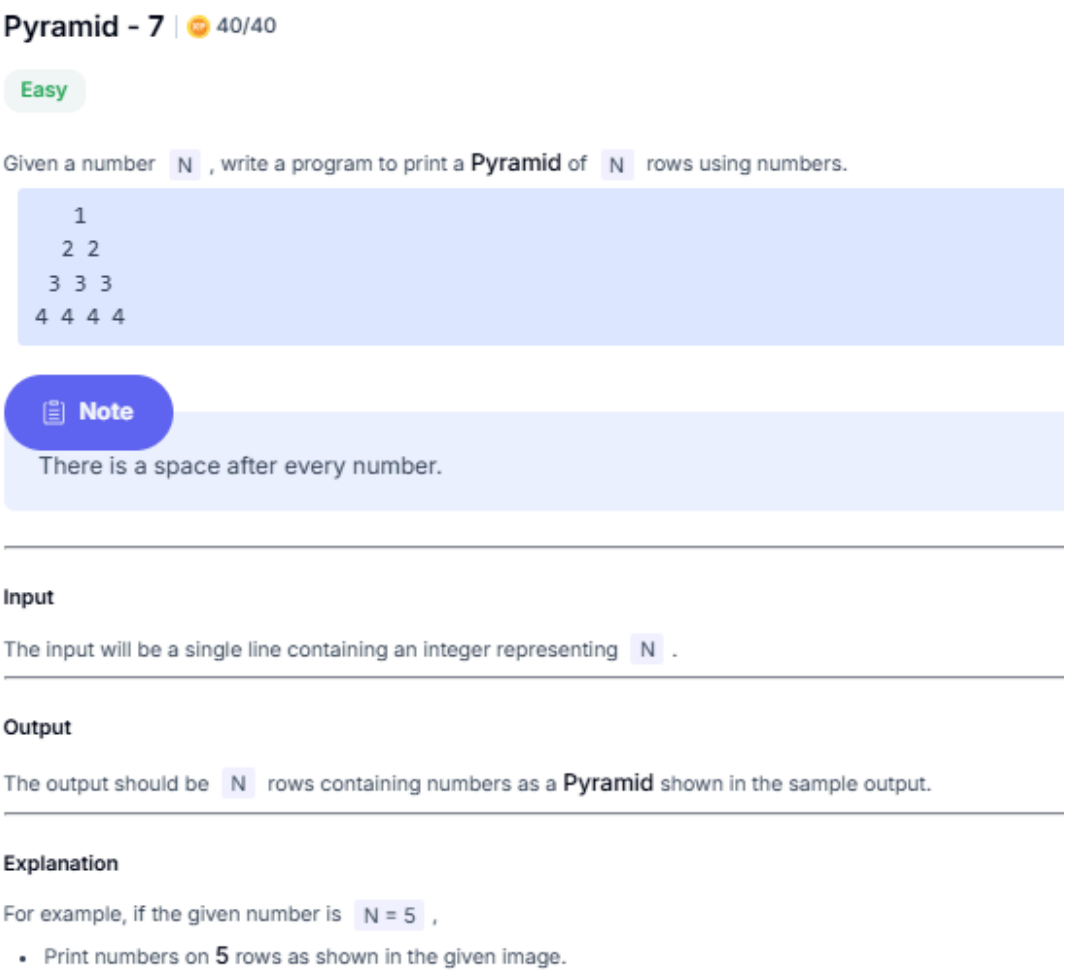
- Print stars (`*`) on **4** rows as shown in the given image.

code:

```
n=int(input())
for i in range(1,n+1):
    row=" "*(n-1)+"* "*i
```

```
    n=n-1
    print(row)
```

7



Pyramid - 7 | 40/40

Easy

Given a number `N` , write a program to print a **Pyramid** of `N` rows using numbers.

```
   1
  2 2
 3 3 3
4 4 4 4
```

Note

There is a space after every number.

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be `N` rows containing numbers as a **Pyramid** shown in the sample output.

Explanation

For example, if the given number is `N = 5` ,

- Print numbers on **5** rows as shown in the given image.

code:

```
n=int(input())
for i in range(1,n+1):
    str_i=str(i)+" "
    space=" "*(n-1)
    row=space+str_i*i
    print(row)
    n=n-1
```

8

Inverted Pyramid - 2 | 35/35

Given a number `N` , write a program to print an **Inverted Pyramid** of `N` rows using stars (`*`).

```
* * * * *
 * * * *
  * * *
   * *
    *
```

Note

There is a space after every star (`*`).

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be `N` rows containing stars (`*`) as an **Inverted Pyramid** shown in the sample output.

Explanation

For example, if the given number is `N = 5` ,

- Print stars (`*`) on **5** rows as shown in the given image.

code:

```
n=int(input())
for i in range(n+1):
    spaces=" "*i
    star="* "*n
    row=spaces+star
    print(row)
    n=n-1
```

9

Inverted Pyramid - 3 | 35/35

Solve

Easy

Given a number `N` , write a program to print an **Inverted Pyramid** of `N` rows using stars (`*`) and pluses (`+`).
The first row of the pyramid contains pluses (`+`), and the remaining rows contain stars (`*`).

```
+ + + +
 * * *
  * *
   *
```

Note

There is a space after every star (`*`) and every plus (`+`).

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be `N` rows containing pluses (`+`) and stars (`*`) as an **Inverted Pyramid** shown in the sample output.

Explanation

For example, if the given number is `N = 4` ,

- Print pluses (`+`) and stars (`*`) on **4** rows as shown in the given image.
  - The first row should contain **4** pluses.
  - The remaining **3** rows should contain stars (`*`).

code:

```
n=int(input())
print("+ "*n)
for i in range(1,n):
        row=(" "*i)+("* "*(n-1))
        print(row)
        n=n-1
```

10

Diamond | 35/35

Given a number `N` , write a program to print a **Diamond** of `(2 * N) - 1` rows using stars (`*`).



Note

There is a space after every star (`*`).

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be `(2 * N) - 1` rows containing stars (`*`) as a **Diamond** shown in the sample output.

Explanation

For example, if the given number is `N = 3` ,

- Print stars (`*`) on **5** (`(2 * 3) - 1`) rows as shown in the given image.

code:

```
n=int(input())
n1=n
for i in range(1,n+1):
    spaces=" "*(n-1)
    star="* "*i
    row=spaces+star
    print(row)
    n=n-1
for i in range(1,n1):
    star="* "*(n1-1)
    spaces=" "*(i)
    row=spaces+star
```

```
    print(row)
    n1=n1-1
```

11

M pattern with * - 2 | 40/40

Easy

Given a number `N` , write a program to print the letter **M** of `N` rows with two **Solid Pyramids** using stars (`*`).

```
  *       *
  * *     * *
  * * *   * * *
* * * * * * * * *
```

Note

There is a space after every star (`*`).

Input

The input will be a single line containing an integer representing `N` .

Output

The output should be `N` rows containing stars (`*`) as the letter **M** shown in the sample output.

Explanation

For example, if the given number is `N = 4` ,

- Print the stars (`*`) on **4** rows as shown in the given image.

code:

```
n=int(input())
for i in range(1,n+1):
    spaces=" "*(n-1)
    star="* "*i
    row=spaces+star+spaces+spaces+star
    print(row)
    n=n-1
```